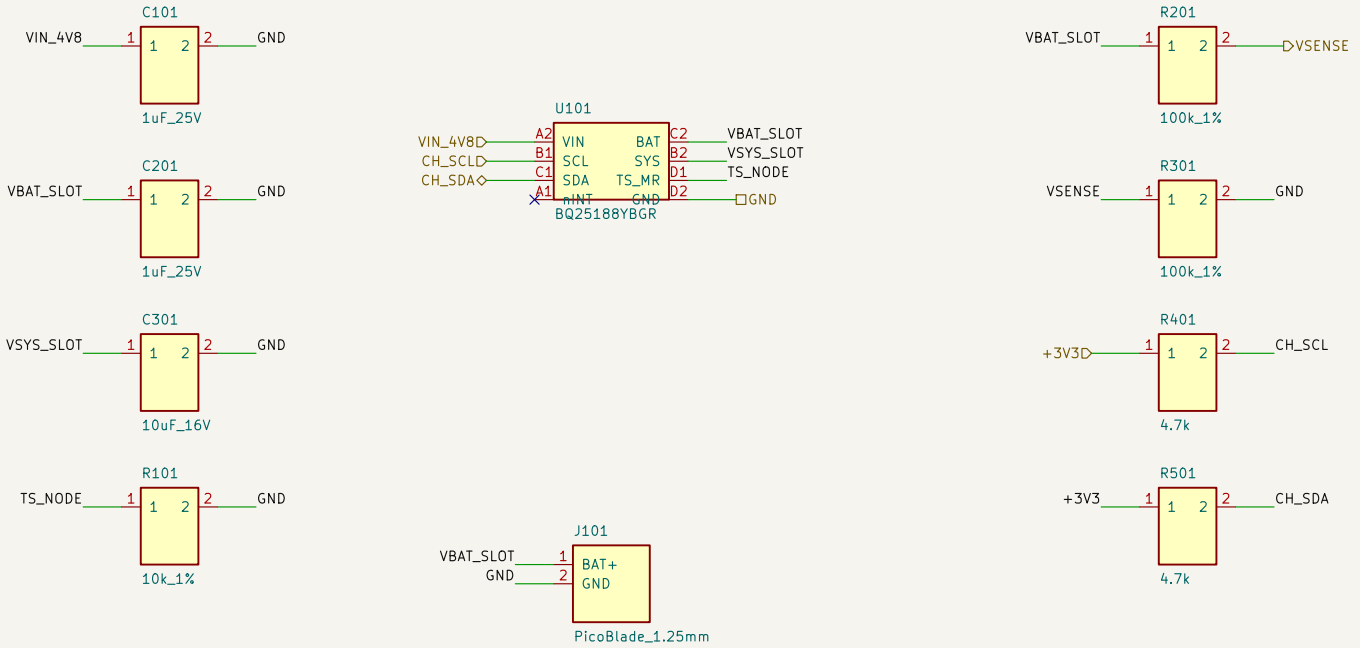
6-SLOT 1S LiPo / LiHV SMART USB-C PD BATTERY CHARGER
- Input: USB-C PD 9V @ 2.5A (22.5W) optimized for Samsung EP-T2510 via AP33771C
- Buck Stage: TI TPS565242DRLR (9V -> 4.80V @ 5A)
- Controller: ESP32-C3-WROOM-02 (BLE 5.0 Android App + Optional 0.96" I2C OLED)
- Channels: 8x Independent TI BQ25188 via TCA9548A I2C Mux + CD74HC4051 Voltage Sense

J1
USB4105-GF-A
U1
AP33771C
U4
ESP32-C3-WROOM-02-N4
D1
Dual_0603_Red_Green
D2
Dual_0603_Red_Green
D3
Dual_0603_Red_Green
D4
Dual_0603_Red_Green
D5
Dual_0603_Red_Green
D6
Dual_0603_Red_Green
Slot_1
Slot_2
Slot_3
Slot_4
Slot_5
Slot_6
R1
82k_1%_9V
R2
100k_1%_Auto
C1
1uF_25V
C2
1uF_25V
R5
10k
C3
22uF_16V
C4
22uF_16V
C6
22uF_16V
C7
22uF_16V
C8
1uF_25V
C9
10uF_16V
Q1
DMN3016LK3
U2
TPS565242DRLR
L1
4.7uH_6A
R3
69.8k_1%
C5
100pF_25V
R4
10k_1%
U3
AP2112K-3.3TRG1
R6
10k
C10
1uF_25V
R7
4.7k
R8
4.7k
R9
10k
R18
10k
R10
10k
SW1
BTN_UP
SW2
BTN_SELECT
SW3
BTN_DOWN
J2
OLED_0.96_I2C
U5
TCA9548APWR
U6
CD74HC4051M96
R11
5.1k_1%
R12
5.1k_1%
R13
5mR_1%
R14
200R
R15
5.1k_1%
C11
100nF_25V
C12
10uF_16V
R16
4.7k_1%
D9
PD_STATUS
C17
22uF_16V
C13
10uF_16V
C14
100nF_25V
C15
100nF_25V
C19
100nF_25V
C16
100nF_25V
C18
100nF_25V
J3
UART
U7
AW9523BTQR

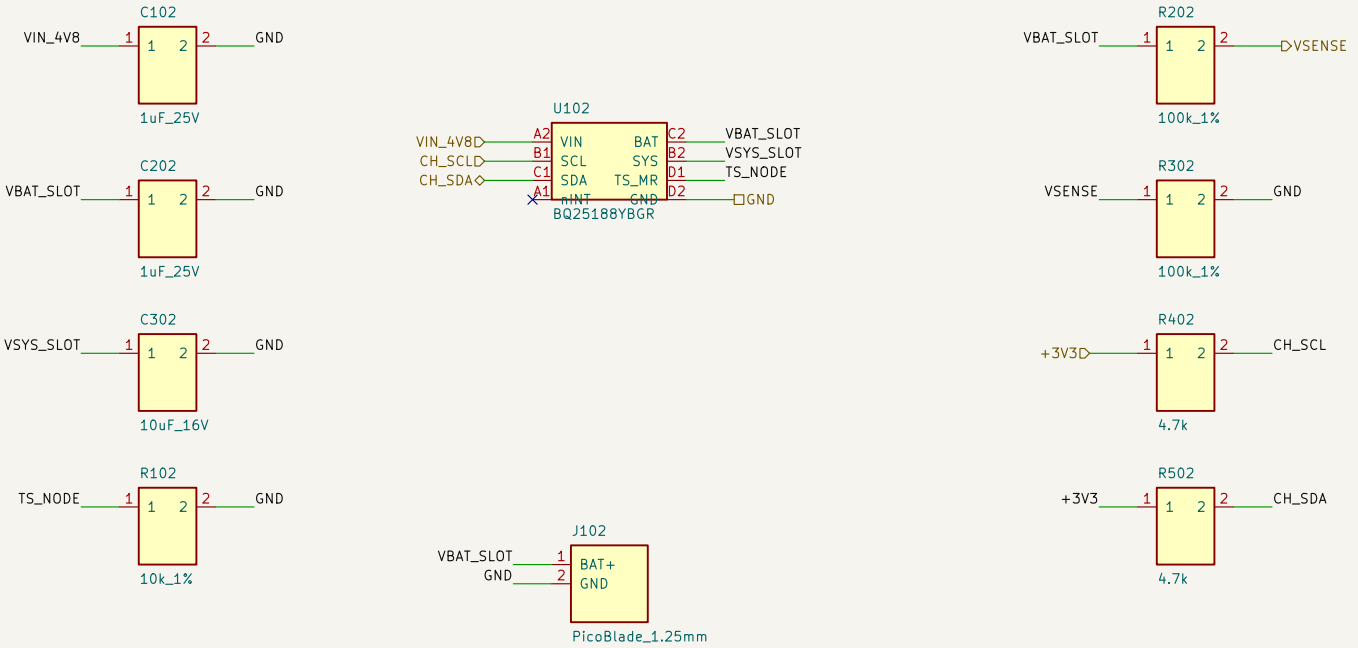
Antigravity Open Hardware
Sheet: /
File: lipo-6slot-charger.kicad_sch
Title: 6-Slot 1S LiPo / LiHV Smart ...ger (ESP32-C3 BLE + BQ2...
Size: A3 | Date: 2025-11-06 | Rev: 1.0
KiCad E.D.A. 10.0.8 | Id: 1/7

PER-SLOT SMART CHARGER CHANNEL (x8 INSTANCES: SLOTS 1-8)
- Charger IC: TI BQ25188YBGR (I2C addr 0x6A via TCA9548A mux)
- Programmable VBAT: 3.60V - 4.65V in 10mV steps (4.20V LiPo / 4.35V LiHV / 3.80V Storage)
- Programmable ICHG: 5mA - 1000mA in 10mA steps (150mA for 150mAh 1C, 500mA for 500mAh 1C)
- Dual Parallel Connectors: Molex PicoBlade 1.25mm (Ultra Micro) + BetaFPV BT2.0



Antigravity Open Hardware		
Sheet: /Slot_1/		
File: charger_slot.kicad_sch		
Title: 1S LiPo/LiHV Programmable Charger Channel (BQ25188 + PicoBlade 1		
Size: A4	Date: 2026-11-06	Rev: 1.0
KiCad E.D.A. 10.0.8		Id: 2/7

PER-SLOT SMART CHARGER CHANNEL (x8 INSTANCES: SLOTS 1-8)
- Charger IC: TI BQ25188YBGR (I2C addr 0x6A via TCA9548A mux)
- Programmable VBAT: 3.60V - 4.65V in 10mV steps (4.20V LiPo / 4.35V LiHV / 3.80V Storage)
- Programmable ICHG: 5mA - 1000mA in 10mA steps (150mA for 150mAh 1C, 500mA for 500mAh 1C)
- Dual Parallel Connectors: Molex PicoBlade 1.25mm (Ultra Micro) + BetaFPV BT2.0



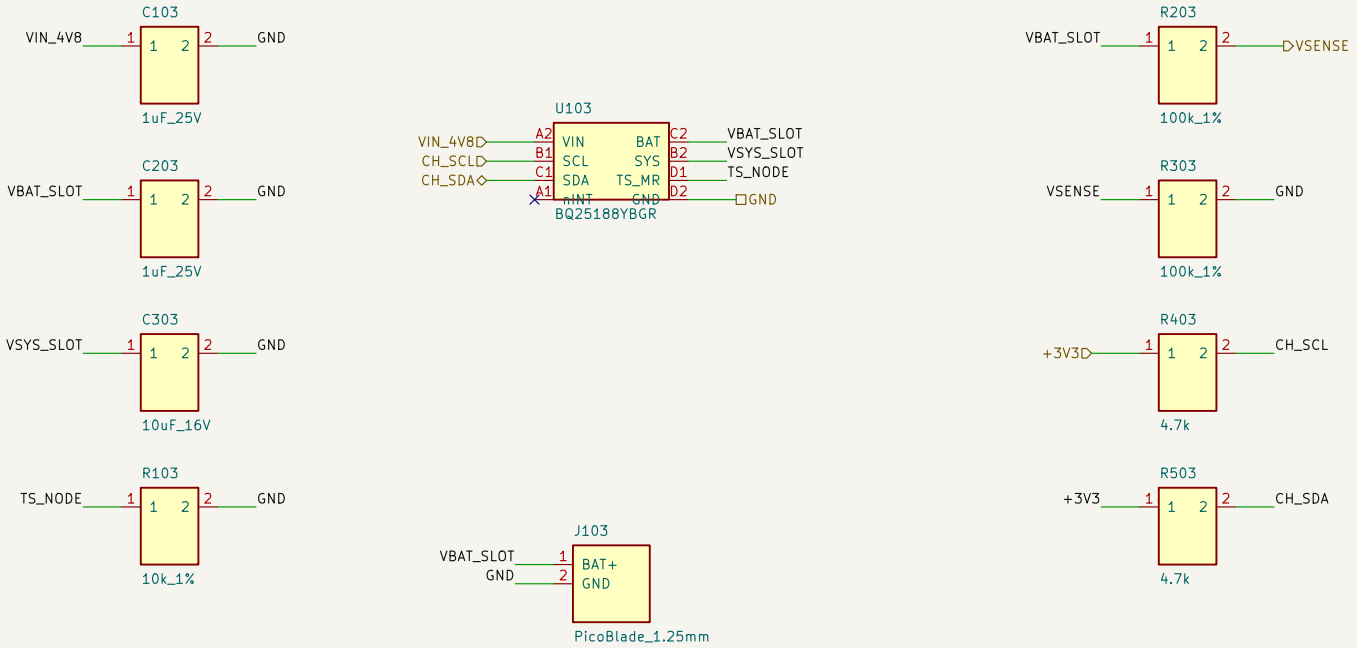
Antigravity Open Hardware

Sheet: /Slot_2/
File: charger_slot.kicad_sch

Title: 1S LiPo/LiHV Programmable Charger Channel (BQ25188 + PicoBlade 1

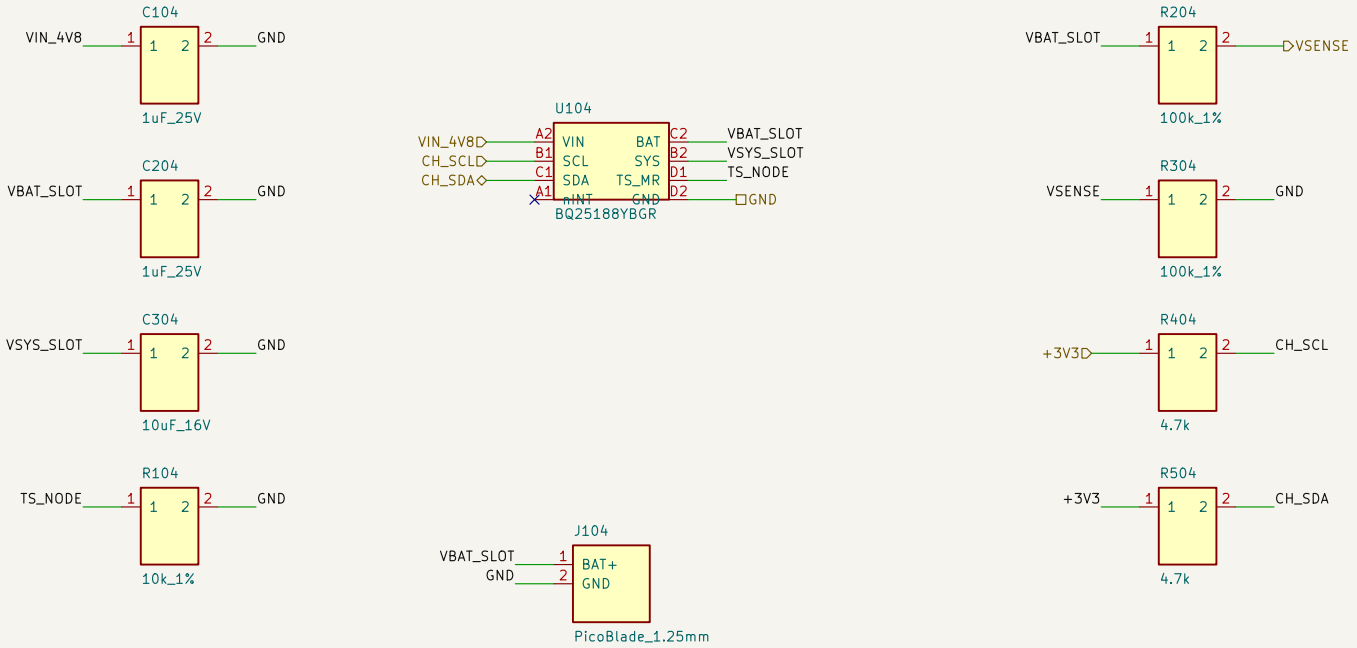
Size: A4 | Date: 2026-11-06 | Rev: 1.0
KiCad E.D.A. 10.0.8 | Id: 3/7

PER-SLOT SMART CHARGER CHANNEL (x8 INSTANCES: SLOTS 1-8)
- Charger IC: TI BQ25188YBGR (I2C addr 0x6A via TCA9548A mux)
- Programmable VBAT: 3.60V - 4.65V in 10mV steps (4.20V LiPo / 4.35V LiHV / 3.80V Storage)
- Programmable ICHG: 5mA - 1000mA in 10mA steps (150mA for 150mAh 1C, 500mA for 500mAh 1C)
- Dual Parallel Connectors: Molex PicoBlade 1.25mm (Ultra Micro) + BetaFPV BT2.0



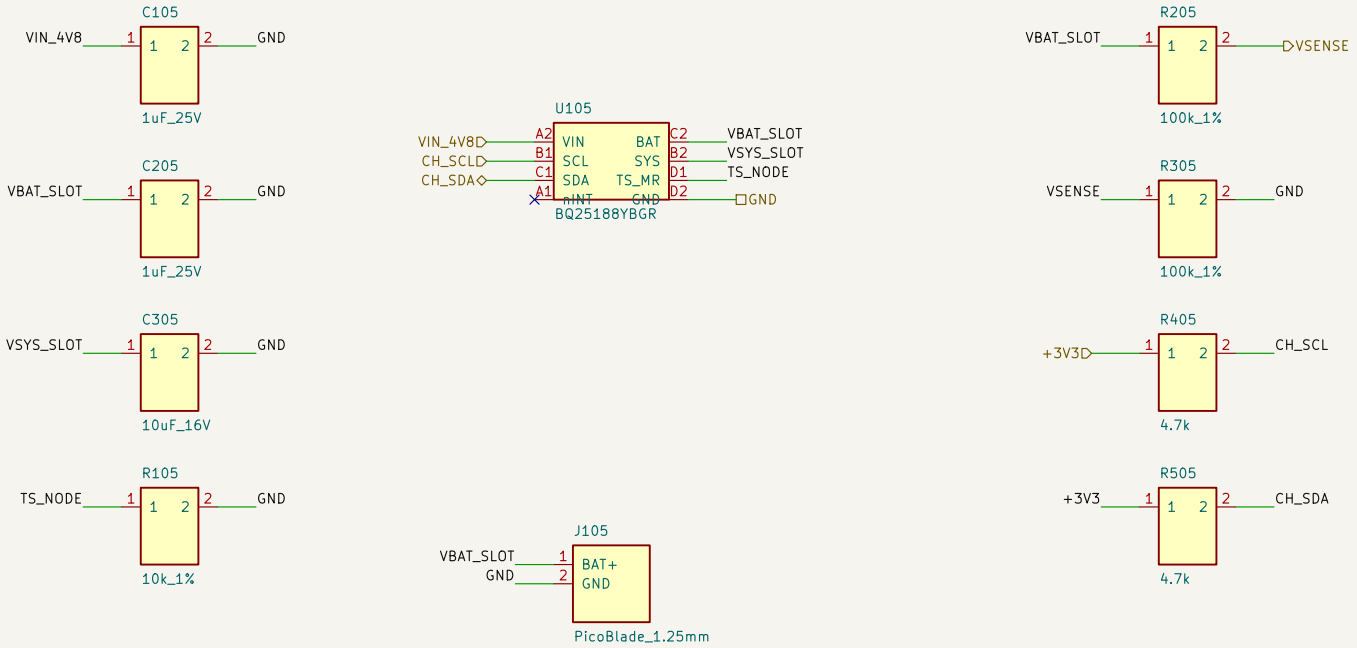
Antigravity Open Hardware		
Sheet: /Slot_3/		
File: charger_slot.kicad_sch		
Title: 1S LiPo/LiHV Programmable Charger Channel (BQ25188 + PicoBlade 1		
Size: A4	Date: 2026-11-06	Rev: 1.0
KiCad E.D.A. 10.0.8		Id: 4/7

PER-SLOT SMART CHARGER CHANNEL (x8 INSTANCES: SLOTS 1-8)
- Charger IC: TI BQ25188YBGR (I2C addr 0x6A via TCA9548A mux)
- Programmable VBAT: 3.60V - 4.65V in 10mV steps (4.20V LiPo / 4.35V LiHV / 3.80V Storage)
- Programmable ICHG: 5mA - 1000mA in 10mA steps (150mA for 150mAh 1C, 500mA for 500mAh 1C)
- Dual Parallel Connectors: Molex PicoBlade 1.25mm (Ultra Micro) + BetaFPV BT2.0



Antigravity Open Hardware		
Sheet: /Slot_4/		
File: charger_slot.kicad_sch		
Title: 1S LiPo/LiHV Programmable Charger Channel (BQ25188 + PicoBlade 1		
Size: A4	Date: 2026-11-06	Rev: 1.0
KiCad E.D.A. 10.0.8		Id: 5/7

PER-SLOT SMART CHARGER CHANNEL (x8 INSTANCES: SLOTS 1-8)
- Charger IC: TI BQ25188YBGR (I2C addr 0x6A via TCA9548A mux)
- Programmable VBAT: 3.60V - 4.65V in 10mV steps (4.20V LiPo / 4.35V LiHV / 3.80V Storage)
- Programmable ICHG: 5mA - 1000mA in 10mA steps (150mA for 150mAh 1C, 500mA for 500mAh 1C)
- Dual Parallel Connectors: Molex PicoBlade 1.25mm (Ultra Micro) + BetaFPV BT2.0



Antigravity Open Hardware

Sheet: /Slot_5/

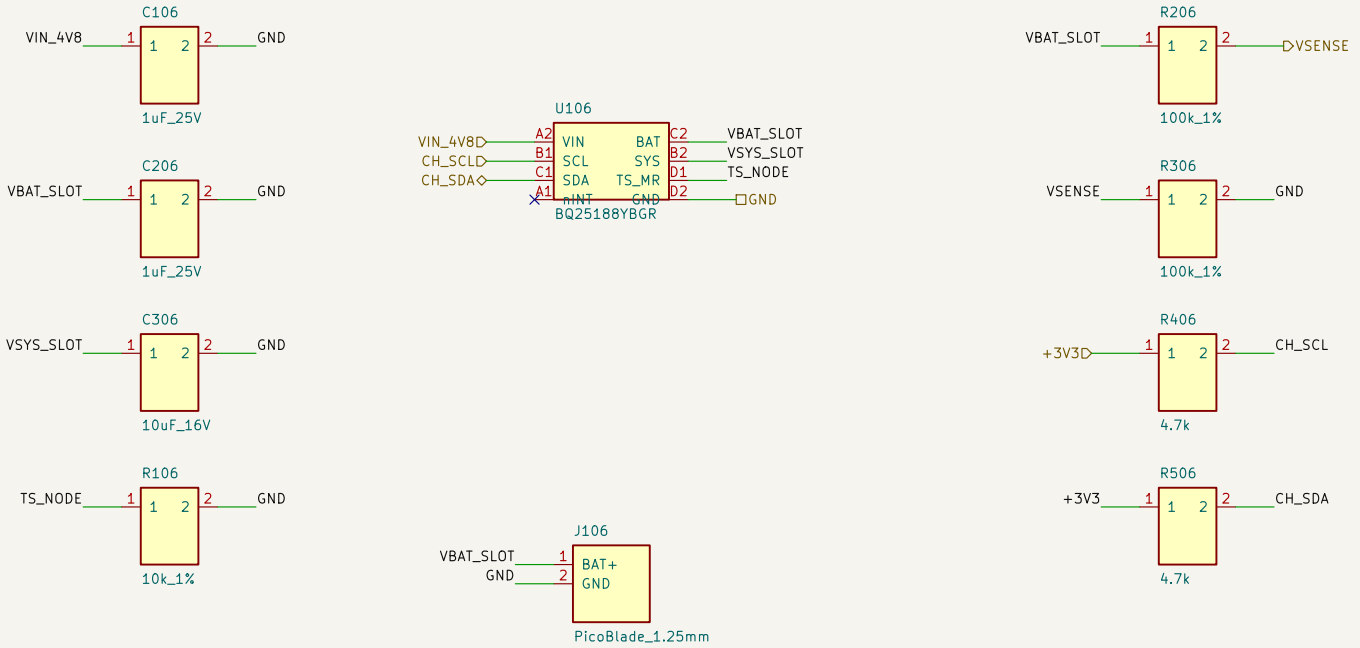
File: charger_slot.kicad_sch

Title: 1S LiPo/LiHV Programmable Charger Channel (BQ25188 + PicoBlade 1

Size: A4 | Date: 2026-11-06 | Rev: 1.0

KiCad E.D.A. 10.0.8 | Id: 6/7

PER-SLOT SMART CHARGER CHANNEL (x8 INSTANCES: SLOTS 1-8)
- Charger IC: TI BQ25188YBGR (I2C addr 0x6A via TCA9548A mux)
- Programmable VBAT: 3.60V - 4.65V in 10mV steps (4.20V LiPo / 4.35V LiHV / 3.80V Storage)
- Programmable ICHG: 5mA - 1000mA in 10mA steps (150mA for 150mAh 1C, 500mA for 500mAh 1C)
- Dual Parallel Connectors: Molex PicoBlade 1.25mm (Ultra Micro) + BetaFPV BT2.0



Antigravity Open Hardware		
Sheet: /Slot_6/		
File: charger_slot.kicad_sch		
Title: 1S LiPo/LiHV Programmable Charger Channel (BQ25188 + PicoBlade 1		
Size: A4	Date: 2026-11-06	Rev: 1.0
KiCad E.D.A. 10.0.8		Id: 7/7